

केंद्रीय विद्यालय संगठन, बेंगलुरु संभाग
KENDRIYA VIDYALAYA SANGATHAN, BENGALURU REGION
 प्रथम प्री बोर्ड परीक्षा 2025-26
FIRST PRE BOARD EXAMINATION 2025-26

CLASS: X

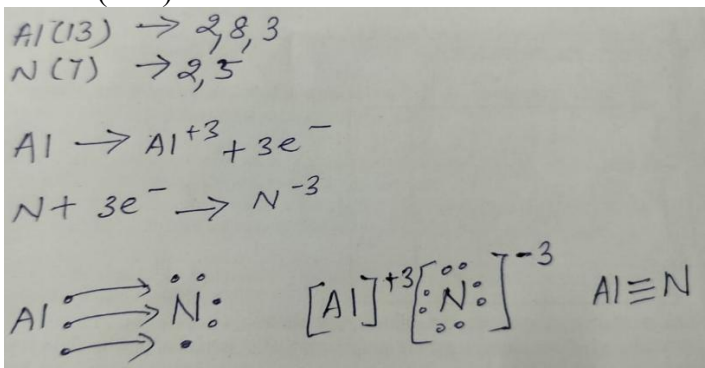
SUBJECT: SCIENCE (Code No. 086)

MAXIMUM MARKS: 80

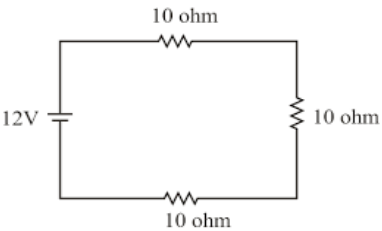
TIME: 3 HRS.

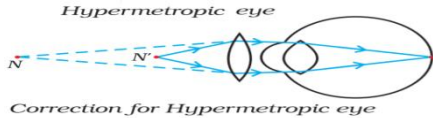
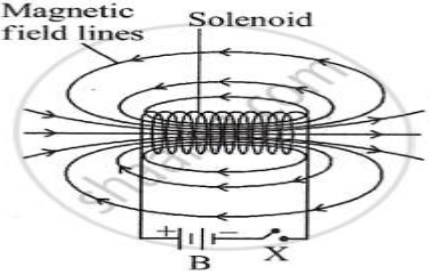
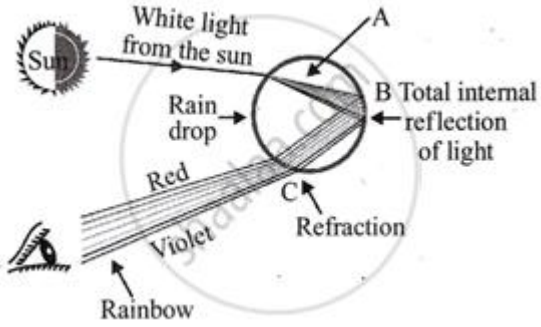
MARKING SCHEME

Section – A				Marks
1.	B.			1
	L (Mouth cavity)	M (Stomach)	N (Small intestine)	
	Amylase	Pepsin	trypsin	
2.	B. Chemotropism			1
3.	D. (ii) and (iv)			1
4.	C. a-iii, b-iv, c-i, d-ii			1
5.	A. 44 + XX			1
6.	C. E/1000			1
7.	B. Sunlight			1
8.	A. Both A and R are true, and R is the correct explanation of A.			1
9.	C. A is true but R is false.			1
10.	During running, muscle cramps occur due to the accumulation of lactic acid. It occurs due to breakdown of glucose in the lack of oxygen (anaerobically). It happens as the body needs a quick supply of energy during heavy physical exercise.			1 1
11.	<i>Students to attempt either option A or B.</i> A. Thyroxin hormone is secreted by thyroid gland. Iodine is essential for the synthesis of thyroxin, so it is important to have iodised salt in our diet for the proper functioning of our body. OR B. Auxin. The tendrils plant are sensitive to the stimuli of touch, when they come in contact with a support (like another plant) uneven distribution of auxin causes growth and coiling of the tendrils around the support here by promoting the elongation of cells helping them to grow and bend towards the support. (thigmotropism -can be awarded ½ mark)			½ + ½ 1 1 1
12.	A – Plumule, B- Cotyledon, C – Radicle B – Cotyledon stores food for the developing embryo.			½ x3 ½
13.	- In the oviduct, sperm encounters the egg and fertilization takes place - The fertilized egg (zygote) starts dividing and form a ball of cells or embryo. - Embryo is implanted in the lining of the uterus, where it continues to grow and develops organs to become a foetus. (flow chat can be awarded marks) - Role of placenta: (a) Provides a large surface area for glucose and oxygen to pass from the mother to the embryo. (b) Waste generated by the embryo will be removed by transferring them into the mother’s blood.			½ ½ ½ ½ ½ + ½ ½
14.	(i) Cereal plants → Human beings (ii) Pesticides being non-biodegradable accumulate progressively at each trophic level leading to Biomagnification.			1 1

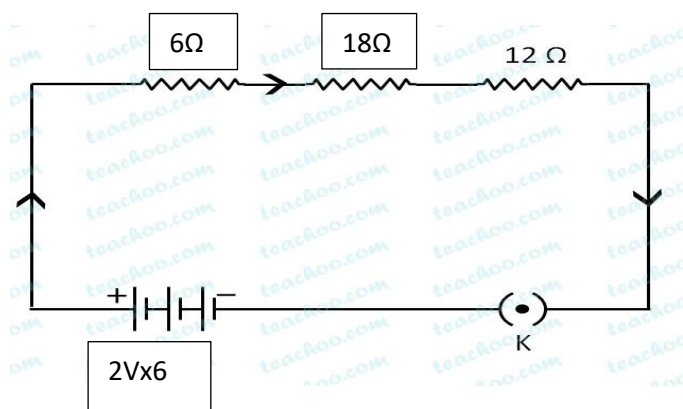
18.	D. (ii) and (iv)	1
19.	D. basic → acidic → basic	1
20.	B. (i), (ii) and (iii)	1
21.	C. Only and (iv)	1
22.	B. Vamika and Vivek are correct.	1
23.	B. 9 and 3	1
24.	B. Both A and R are true, and R is not the correct explanation of A.	1
25.	A – Sodium (Na) or Potassium (K) B – Calcium (Ca) or Magnesium (Mg), because the bubbles of H ₂ gas formed get stick to the surface of the metal. C – Iron (Fe), Aluminium (Al) or Zinc (Zn) D – Gold (Au) or Silver (Ag)	0.5 * 4
26.	Students to attempt either option A or B. A. (i) Aluminium can loose three electrons from its outermost shell, while nitrogen can gain three electrons in its outermost shell to complete their octet. Formation of aluminium nitride (AlN):  Al≡N (ii) Ionic compounds are solids and are generally brittle because the strong electrostatic forces of attraction between the oppositely charged ions form a rigid structure. When pressure is applied, like charged ions repel each other, weakening the crystal structure and causing it to break into pieces OR B. (i) Anode: Thick impure block of silver Cathode: Thin strip of pure silver (ii) Electrolyte: An aqueous solution of a silver salt such as AgNO₃, AgCl, etc. (iii) We get pure silver at the cathode.	½ ½ ½ ½ 1 1 1 1
27.	(i) 'X' is ethanol C₂H₅OH (ii) The chemical equations for the above mentioned reactions are as follows: $2\text{C}_2\text{H}_5\text{OH} + 2\text{Na} \rightarrow 2\text{C}_2\text{H}_5\text{O}^-\text{Na}^+ + \text{H}_2\uparrow$ $\text{C}_2\text{H}_5\text{OH} \xrightarrow[445\text{ K}]{\text{H}_2\text{SO}_4} \text{CH}_2 = \text{CH}_2 + \text{H}_2\text{O}$ Concentrated sulphuric acid acts as a dehydrating agent in the second reaction.	1 1 1
28.	A. (i) Group A and Group B and (ii) Group C B. By using universal indicator and pH paper Attempt either subpart C or D. C. Copper vessel is tarnished due to the formation of basic copper carbonate. Lemon juice contains citric acid which can react with the basic copper substance to form a salt which gets washed away. OR D. For gardening and farming, the best crops are usually obtained with neutral or very slightly acidic soil (pH – 6.5 to 7.0). Below pH 6.5, the soil is not suitable for normal growth of plants, particularly vegetables.	1 1 2
29.	Attempt either option A or B. A.	

	(i) Thermal decomposition reaction: In a decomposition reaction, a single substance decomposes to give two or more simpler substances. Specifically, the green reaction is a thermal decomposition reaction since it uses heat as its source of energy.	1
	(ii) Nitrogen dioxide (NO ₂) gas is liberated which is brown in colour.	1
	$2\text{Pb}(\text{NO}_3)_2 (\text{s}) \xrightarrow{\text{Heat}} 2\text{PbO} (\text{s}) + 4\text{NO}_2 (\text{g}) + \text{O}_2 (\text{g})$ Lead nitrate Lead oxide Nitrogen dioxide (Oxygen) (pale yellow) (Brown)	1
	(iv) Lead oxide remains as a residue in the test tube. Dissolve this residue in water and test the solution using red litmus paper. The colour of the litmus paper changes to blue indicating the basic nature of lead oxide.	2
	OR	
	B. (i) A decomposition reaction is a type of chemical reaction where a single compound breaks down into two or more simpler substances.	1
	(a) Electrolysis of Water:- Reaction: $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$ Energy: Electrical energy Explanation: Water is decomposed into hydrogen and oxygen gases when an electric current is passed through it.	1
	(b) Blackening of Silver Bromide:- Reaction: $2\text{AgBr} \rightarrow 2\text{Ag} + \text{Br}_2$ Energy: Light energy Explanation: Silver bromide decomposes into silver and bromine when exposed to sunlight, causing the silver bromide to turn black. Type of Energy Involved:- Electrolysis of Water: Electrical energy Blackening of Silver Bromide: Light energy	1
	(ii) Opposite Reactions: Calcium Oxide and Calcium Hydroxide Statement: The type of reactions in which calcium oxide is formed and calcium hydroxide is formed are opposite reactions to each other. Justification: (a) Formation of Calcium Oxide:- Reaction: $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ (decomposition reaction) - Explanation: Calcium carbonate decomposes into calcium oxide and carbon dioxide when heated. (b) Formation of Calcium Hydroxide:- Reaction: $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca}(\text{OH})_2$ (combination reaction) Explanation: Calcium oxide reacts with water to form calcium hydroxide. Opposite Reactions: These two reactions are opposite because one involves the decomposition of calcium carbonate to form calcium oxide, while the other involves the combination of calcium oxide with water to form calcium hydroxide.	2
SECTION – C		
30.	B. (i) and (iv) only	1
31.	A. a-iv, b-iii, c-ii, d-i	1
32.	D. A is false but R is true.	1
33.	Absolute Refractive Index: The absolute refractive index of a medium is defined as the ratio of the speed of light in vacuum to the speed of light in the medium. Formula: $n = c / v$ where: n = absolute refractive index c = speed of light in vacuum (3×10^8 m/s) v = speed of light in the medium Given Values: $n = 1.5$ $c = 3 \times 10^8$ m/s Speed of Light in Glass: $v = c / n$	2

	$= (3 \times 10^8 \text{ m/s}) / 1.5$ $= 2 \times 10^8 \text{ m/s}$ The speed of light in glass is $2 \times 10^8 \text{ m/s}$.	
34.	Students to attempt either option A or B. A. Applications of Joule's Heating in Domestic Electric Circuit (i) Applications: 1. Electric Heating Appliances: Joule's heating is used in electric heaters, toasters, and geysers to convert electrical energy into heat energy. 2. Incandescent Bulbs: Joule's heating is used in incandescent bulbs to heat the filament until it glows, producing light. (ii) Relationship between Commercial Unit and SI Unit of Electric Energy:- Commercial Unit: 1 kilowatt-hour (kWh) SI Unit: 1 joule (J) Establishing the Relationship: $1 \text{ kWh} = 1 \text{ kW} \times 1 \text{ h}$ $= 1000 \text{ W} \times 3600 \text{ s}$ $= 3,600,000 \text{ J}$ $= 3.6 \times 10^6 \text{ J}$ Therefore, $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$. OR B. Minimum Heat Production Circuit To minimize heat production, connect the resistors in series. Joule's law states that heat energy is directly proportional to resistance when the current is kept constant, but inversely proportional when the voltage is kept constant. $=$  Total Resistance (R_t) = $R_1 + R_2 + R_3 = 10 + 10 + 10$ 30Ω This series connection maximizes total resistance, minimizing current and heat production. The voltage (4V) is kept constant because each resistor is of 10 ohms	1 1 $\frac{1}{2}$ $1 \frac{1}{2}$
35.	(i) Calculations for Concave Lens Given Values: Object height (h) = 5 cm Object distance (u) = -20 cm (negative sign indicates the object is in front of the lens) Focal length (f) = -18 cm (negative sign indicates a concave lens) (a) Image Distance (v): Using the lens formula: $1/f = 1/v - 1/u$ $1/(-18) = 1/v - 1/(-20)$ $-1/18 = 1/v + 1/20$ $1/v = -1/18 - 1/20$ $1/v = (-10 - 9) / 180$ $1/v = -19 / 180$ $v = -180 / 19$ $v \approx -9.47 \text{ cm}$ (b) Magnification (m): $m = v / u$ $= (-9.47) / (-20)$ $= 0.4735$ (ii) Comparison of Magnification Concave Lens:- Always forms virtual, erect, and diminished images. - Magnification is always less than 1. Convex Lens (forming virtual image):- Forms virtual, erect, and magnified images when the object is between the lens and the focal point. - Magnification is greater than 1.	2 1

	Comparison: The magnification of a concave lens is always less than 1, while the magnification of a convex lens forming a virtual image is greater than 1.	
36.	(i) Defect of Vision -The defect of vision represented in the diagram is Hypermetropia (Long-sightedness). Reason: In hypermetropia, the eyeball is too short or the cornea is too flat, causing light rays to focus behind the retina instead of directly on it. (ii) Causes of Hypermetropia: 1. Shortened eyeball: The eyeball is shorter than normal, causing light rays to focus behind the retina. 2. Elongation of the eye lens . (iii) Correction : Convex lens 	1 1 1
37.	A solenoid is a coil of wire wrapped in a cylinder that acts as an electromagnet when current flows through it. Its magnetic field resembles that of a bar magnet, with uniform, parallel field lines inside and closed loops outside. To magnetize a soft iron piece, it is placed inside the solenoid; the solenoid's strong magnetic field aligns the magnetic domains within the iron, turning it into an electromagnet. 	2 1
38.	A. Another instance of observing a spectrum of white light is during a rainbow formation in the sky after a rain shower. B. In the above case, white light gets dispersed into its constituent colors, forming a spectrum. This occurs because each color of light has a slightly different wavelength and is refracted at a slightly different angle as it passes through the prism. C. 1. Sunlight: Rainbows require sunlight to occur. 2. Water Droplets: Rainbows require water droplets in the air, such as after a rain shower or near a waterfall. OR 	1 1 2

39.

Attempt either option A or B.**A** (i) $1/R = 1/6 + 1/12 + 1/18 = 6/18$, $R = 18/3 = 3 \Omega$ (ii) 12Ω and 6Ω in parallel = 4Ω and this in series with 18Ω $4 + 18 = 22 \Omega$ +Diagram(iii) $I = Q/t$ $50 C / 10 s = 5 A$ **OR**Total Voltage: $V = 6 \text{ cells} \times 2V/\text{cell} = 12V$ Total Resistance: $R = 6\Omega + 12\Omega + 18\Omega = 36\Omega$ (i) Electric Current: $I = V / R = 12V / 36\Omega = 1/3 A = 0.33 A$ (ii) Potential Difference across 18Ω : $V = I \times R = 0.33 A \times 18\Omega = (5.94) \text{ or } 6V \text{ approx}$ (iii) Electric Power: $P = I^2 \times R = (0.33 A)^2 \times 18\Omega = 1.96 W$

The answers are:

(i) $0.33 A$ (ii) $6 V$ (iii) $1.96 W$
